Supply Chain & Logistics Performance Analysis

End-to-End Problem, Workflow, Insights & Recommendations

Business Problem

The business needs a consolidated way to analyze product demand, revenue, inventory, suppliers, shipping, manufacturing, and quality performance.

End-to-End Workflow

Raw Dataset → Python data preparation and validation → Cleaned dataset → MySQL business analysis → Power BI dashboard → Business insights → Recommended actions.

Technology Solution

Python and Pandas were used for data preparation and validation. MySQL and SQL were used to analyze business questions. Power BI and DAX were used to build KPIs, interactive visuals, and the final four-page dashboard.

Dashboard Solution

The solution contains Executive Overview, Inventory & Supply Chain, Manufacturing & Quality, and Insights & Actions pages.

Key Insights

Skincare generated the highest revenue at approximately ■0.24M and the highest product sales at approximately 21K units. Cosmetics had the highest average stock level. Haircare had the highest average defect rate at approximately 2.48%. Supplier lead times varied, and Skincare had the highest production volume at approximately 24K units.

Recommended Actions

Improve quality control for Haircare; monitor inventory for high-demand Skincare products; review suppliers with longer lead times; analyze shipping costs for cost-effective transportation choices; and align production planning with product demand.

Business Value

The project turns operational data into a single decision-support dashboard, helping stakeholders identify performance patterns and prioritize supply chain improvements.